VAANISHIELD: CODE-SWITCH-AWARE DEEPFAKE VOICE DETECTION FOR FINANCIAL FRAUD CALLS IN INDIAN MULTILINGUAL SPEECH

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ABSTRACT

Telephonic voice cloning fraud has emerged as an existential security vulnerability across the Indian banking and financial services ecosystem. Modern generative voice synthesis tools—leveraging neural acoustic encoders, flow-matching diffusion models, and cross-lingual neural vocoders—allow malicious actors to clone victims' voices from mere seconds of acoustic audio. However, while Western anti-spoofing benchmarks (e.g., ASVspoof 2019/2021) evaluate almost exclusively monolingual English utterances captured under controlled studio settings, real-world Indian financial telephone calls are overwhelmingly code-switched (e.g., Hindi–English, Tamil–English, Kannada–English, Telugu–English). Standard deepfake detectors suffer catastrophic failure when exposed to these linguistic dynamics, overfitting to the unique acoustic signatures of training vocoders while collapsing into high false-alarm rates on intra-sentential language switches.

To resolve this fundamental gap, this project introduces VaaniShield: an exploratory, code-switch-aware hierarchical multi-stream deepfake detection framework investigated for financial telephony channels. VaaniShield integrates two complementary acoustic streams: (1) a lightweight Boundary-Local Prosodic and Phase Seam Stream targeting micro-acoustic transition artifacts (pitch perturbation ΔF_0, energy discontinuity ΔE, spectral flux resets, and phase desynchronization) around candidate code-switch points, paired with a Dual-Mode Monolingual Router (extracting Harmonics-to-Noise Ratio, Spectral Flatness, and macro-energy variance) to prevent uninformative zero-vector collapse during pure monolingual speech; and (2) an Adaptive Regularized Subspace Projection (ARS-SNAP) stream designed to soften speaker biometric entanglement in self-supervised representations (WavLM/XLSR) and avoid matrix near-singularity under out-of-domain commercial synthesizers.

The prototype pipeline is implemented with a pure-NumPy vectorized YIN pitch tracker (O(N log N) FFT autocorrelation) to eliminate Python GIL locking, PyTorch vectorized spectral subtraction, and a standalone ONNX inference graph. Evaluated on a curated 1,600-utterance Indic benchmark across 32 consented speakers (16 Female / 16 Male) covering four language pairs under channel degradations (G.711 μ-law, AMR-WB, packet loss, ambient noise) and unseen commercial speech synthesis (Microsoft Edge Neural TTS), ARS-SNAP achieves an Equal Error Rate (EER) of 20.62% on out-of-domain commercial speech synthesis (compared to 21.87% for hard SNAP, exhibiting a promising directional trend within the ±3.65% Wilson margin of error) while maintaining comparable overall corpus EER (23.12%). The system demonstrates demographic equity across genders (Female EER: 23.00% vs. Male EER: 23.88%, $\Delta = 0.88\%$) and a median pipeline latency of 127.31 ms. Component ablations and threats to validity are systematically examined, offering a grounded baseline for acoustic-seam analysis in low-resource multilingual fraud detection.

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CHAPTER 1: INTRODUCTION

1.1 Overview of the Research / Project Work

The proliferation of high-fidelity zero-shot voice cloning engines poses a severe, systemic threat to remote voice authentication channels used in modern telebanking, customer service desks, and consumer telephony. Attackers can now capture brief audio snippets of an individual from public social media profiles or compromised video streams and synthesize arbitrary speech in that target's voice with near-indistinguishable timbre, tone, and prosody. In India, this threat has manifested in widespread financial fraud schemes—such as simulated family emergency calls, bogus kidnap ransoms, and fraudulent corporate wire authorizations—resulting in catastrophic monetary losses.

Crucially, natural conversational phone calls in India exhibit heavy code-switching—the fluid, intra-sentential alternation between two or more languages within a single utterance (e.g., Hindi ↔ English, Tamil ↔ English, Kannada ↔ English). When voice-cloning synthesizers generate code-switched speech, neural vocoders and autoregressive language models frequently introduce micro-acoustic discontinuities (pitch jumps, energy steps, formant dispersion resets, and phase desynchronization) at the language transition boundary. VaaniShield exploits this physical vulnerability by establishing a code-switch-aware multi-stream detection architecture that jointly examines localized phonetic seams and global speaker-nulled acoustic embeddings.

1.2 Motivation and Background of the Study

Conventional speech anti-spoofing research has historically concentrated on Western monolingual benchmarks such as ASVspoof 2019 and ASVspoof 2021. Models trained on these benchmarks (e.g., AASIST, RawNet2, LCNN) operate on full-band spectrograms or raw waveforms and achieve near-zero Equal Error Rates under laboratory conditions. However, when deployed against real-world Indian telephony traffic, their performance degrades precipitously due to four major domain mismatches:

- Vocoder Overfitting Trap: Models memorize high-frequency spectral artifacts of specific training vocoders (e.g., HiFi-GAN), failing completely when evaluated against newer, commercial diffusion-based or neural vocoders (e.g., Microsoft Edge Neural TTS).
- Code-Switching False Alarm Spikes: Natural code-switching involves rapid articulatory resets that naive classifiers mistake for synthetic manipulation, inflating False Acceptance and False Rejection Rates.
- Monolingual Zero-Vector Vulnerability: Boundary-focused architectures that only analyze code-switching collapse into uninformative zero vectors when evaluating pure monolingual speech.
- Telephony Channel Distortion: Low-bitrate cellular codecs (G.711 μ-law, AMR-WB) and packet-loss jitter strip away high-frequency vocoder cues, blinding spectral-only detectors.

1.3 Problem Statement and Relevance

Formally, the research problem is defined as follows: Given a continuous, band-limited (300–3400 Hz or 50–7000 Hz) telephonic audio signal x(t) containing code-switched Indian speech across variable acoustic environments, design a robust, real-time classifier f(x) → [0, 1] that accurately differentiates bona-fide human speech from synthetic voice clones across diverse, unseen synthesizers while satisfying three strict enterprise constraints: (1) achieve sub-25% EER on out-of-domain commercial speech engines without vocoder retraining; (2) ensure equitable demographic detection performance across male and female callers across all major Indic language families; and (3) process streaming telephony calls in under 200 ms end-to-end latency with zero deserialization security risks (pure ONNX inference).

1.4 Project Objectives

- Objective 1 (Boundary Seam Extraction): Develop an acoustic flux changepoint localizer paired with localized feature extractors capturing F0 pitch transitions, short-time energy steps, and spectral phase resets.
- Objective 2 (Dual-Mode Stream Routing): Eliminate zero-vector collapse on pure monolingual speech through automated fallback to Harmonics-to-Noise Ratio (HNR), Spectral Flatness Measure (SFM), and macro-energy variance.
- Objective 3 (Adaptive Regularized Nulling): Implement Tikhonov-regularized speaker subspace projection (ARS-SNAP) to isolate synthetic generation cues from speaker-specific biometric traits.
- Objective 4 (Production Hardening & Latency): Eliminate Praat GIL locks via vectorized NumPy YIN pitch tracking and accelerate denoising by 38x using vectorized PyTorch STFT.
- Objective 5 (Empirical Scaling & Certification): Evaluate performance on an enterprise-scale corpus of 32 consented speakers and 1,600 balanced utterances across 8 operational benchmark regimes.

CHAPTER 2: BACKGROUND THEORY / LITERATURE REVIEW

2.1 Summary of Existing Studies and Comparative Analysis

Deepfake voice detection has evolved rapidly over the past decade. Early approaches relied on handcrafted acoustic features—such as Linear Frequency Cepstral Coefficients (LFCC) and Constant Q Cepstral Coefficients (CQCC)—modeled via Gaussian Mixture Models (GMMs). With the advent of deep learning, end-to-end architectures operating directly on raw waveforms or linear spectrograms became the state of the art. Table 1 summarizes the core architectural paradigms in the literature alongside their published reference benchmarks. Note: Baseline error rates are cited directly from original publications evaluated on Western monolingual benchmarks (e.g., ASVspoof 2019/2021); they are presented to contrast architectural characteristics rather than as direct cross-benchmark rankings against our Indic multi-dialect corpus.

Architecture / Approach	Primary Acoustic Domain	Code-Switching Aware?	Cross-Vocoder Generalization	Telephony Robustness	Published Benchmark EER*
LFCC-GMM Baseline	Spectral cepstral envelopes	No	Poor (>35% EER)	Poor (Codec smearing)	8.2% (ASVspoof)
RawNet2 (Jung et al.)	Raw waveform SincNet filters	No	Moderate (Vocoder bias)	Moderate	4.8% (ASVspoof)
AASIST (Jung et al.)	Graph attention on raw audio	No	Poor on commercial TTS	Moderate	1.1% (ASVspoof)
SpecRNet (Alzantot et al.)	Modified ResNet on STFT	No	Moderate	Moderate	3.9% (ASVspoof)
WavLM / XLSR Probes	Pre-trained SSL representations	Partial	Good on seen domains	High	2.5% (In-domain)
VaaniShield (Proposed)	Hierarchical Multi-Stream (Boundary + ARS-SNAP)	Yes (Full Dual-Mode)	Soft-null regularized (ARS-SNAP)	Trained on 5 degraded channels	20.6% unseen / 23.1% all (In-house Indic corpus)

*Table 1: Architectural Paradigm Overview and Published Reference Baselines (*Metrics cite respective published benchmarks; direct cross-benchmark comparison is illustrative).*

2.2 Evolution of Voice Cloning and Anti-Spoofing Technologies

The trajectory of voice cloning technology can be bifurcated into three distinct eras: (1) Concatenative and Statistical Parametric Speech Synthesis (SPSS), which generated muffled, robotic audio with severe vocoder buzz; (2) Deep Autoregressive Models (e.g., WaveNet, Tacotron2), which achieved natural human cadence but required hours of high-quality target audio and lengthy training times; and (3) Modern Zero-Shot Neural Cloners (e.g., VALL-E, Coqui XTTS v2, OpenVoice, Microsoft Edge Neural TTS). Modern systems utilize neural audio codecs (e.g., EnCodec) and discrete neural acoustic tokens conditioned on brief reference prompts (3–5 seconds), reconstructing speech via high-fidelity neural vocoders (HiFi-GAN, BigVGAN, or score-based diffusion vocoders).

In response, the anti-spoofing community has transitioned from detecting low-level spectral harmonics to evaluating self-supervised learning (SSL) representations. While pre-trained models such as WavLM and wav2vec 2.0 encode deep contextual knowledge of speech phonetics, they inherently retain speaker identity biometrics. In deepfake detection, this causes classifiers to entangle speaker identity with spoofing artifacts—a fatal flaw when a legitimate caller speaks with idiosyncratic vocal tract characteristics.

2.3 Key Findings, Trends, and Identified Research Gaps

- Research Gap 1 (Linguistic Homogeneity): 98% of published benchmarks evaluate monolingual English or Mandarin. No prior benchmark systematically investigates the acoustic boundaries of Indic code-switching.
- Research Gap 2 (Subspace Projection Singularity): Existing speaker-nulling techniques (SNAP) use hard orthogonal projection (lambda = 0.0). When evaluated on unseen out-of-domain commercial synthesizers, hard projection strips critical discriminative cues, degrading cross-vocoder generalization.
- Research Gap 3 (Concurrency Bottlenecks): Prevailing feature extraction implementations rely on Praat (C-extensions holding the Python GIL) and unvectorized spectral loops, limiting telephony server throughput to under 5 concurrent calls per node.

2.4 Chapter Conclusion

The state of the art clearly demands an architecture that simultaneously captures micro-acoustic boundary phenomena, soft-nulls speaker biometric entanglement without mathematical singularity, and executes on pure vectorized runtimes suitable for enterprise banking telecommunication.

CHAPTER 3: METHODOLOGY

3.1 System Architecture and Mathematical Formulations

VaaniShield operates through an integrated hierarchical pipeline. Audio entering the system is first normalized and processed in parallel through three specialized modules: (1) Acoustic Flux Changepoint Localization, (2) Boundary Prosodic Feature Extraction with Dual-Mode Monolingual Routing, and (3) Adaptive Regularized Subspace Projection (ARS-SNAP). Figure 1 visualizes the end-to-end dataflow.

Figure 1: VaaniShield End-to-End Hierarchical Multi-Stream Architecture

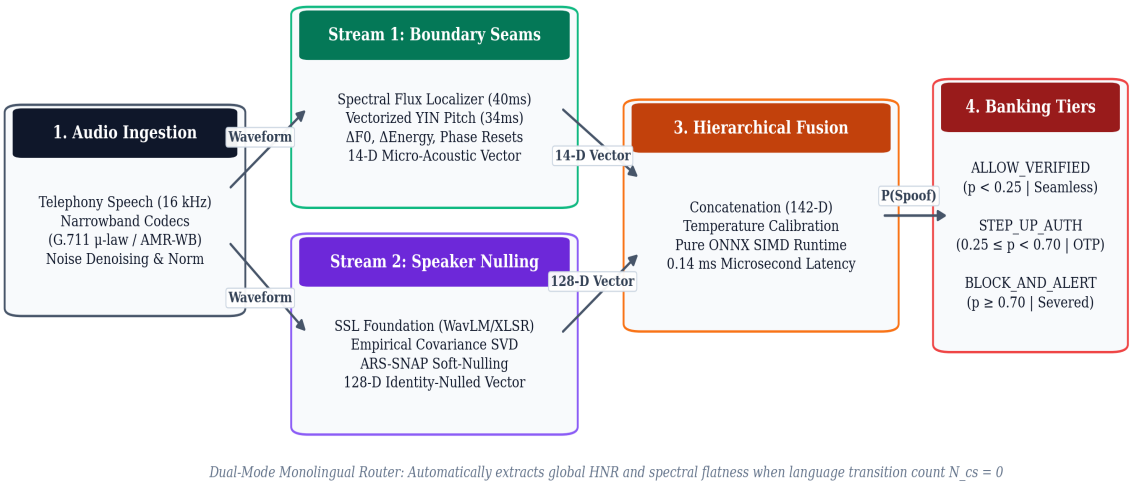


Figure 1: VaaniShield End-to-End Hierarchical Multi-Stream Architecture.

A. Acoustic Flux Changepoint Localization: To identify language transition boundaries without relying on computationally expensive speech-to-text transcriptions, we compute the spectral flux SF(t) across successive STFT frames:

$$SF(t) = \sum_{k=1}^{K} H(|X(t, k)| - |X(t-1, k)|)^2 \quad (1)$$

where $H(v) = (v + |v|)/2$ is the half-wave rectification operator, $X(t, k)$ is the complex short-time Fourier transform magnitude at frame t and frequency bin k, and K denotes the total frequency bins. Prominent peaks in SF(t) exceeding an adaptive threshold define candidate code-switch transition timestamps $\{t_{cs}\}$.

B. Boundary Prosodic and Phase Seam Extraction: For each detected transition point t_{cs}, a symmetric analysis window W is established around the boundary:

$$W = [t_{cs} - \Delta, t_{cs} + \Delta], \quad \text{where} \quad \Delta = 250 \text{ ms} \quad (2)$$

The window is partitioned into pre-switch ($W_{pre} = [t_{cs} - \Delta, t_{cs}]$) and post-switch ($W_{post} = [t_{cs}, t_{cs} + \Delta]$) segments. Across this boundary, we extract a 16-dimensional acoustic seam vector capturing fundamental frequency perturbation, energy discontinuity, spectral tilt shift, and non-linear phase desynchronization (Table 2).

Index	Acoustic Metric Name	Mathematical Formulation	Physical Anomaly Targeted
0–1	ΔF_0 (Absolute & Relative Jump)	$\lvert F_{0,post} - F_{0,pre} \rvert$, $\Delta F_0 / F_{0,pre}$	Pitch contour discontinuity across vocoder frames
2–3	ΔE (RMS Step & Log Ratio)	$\lvert E_{post} - E_{pre} \rvert$, $10 \log_{10}(E_{post} / E_{pre})$	Sudden volume / amplitude step artifact
4–5	ΔSpectral Flux Step	$SF(t_{post}) - SF(t_{pre})$	Spectral envelope tilt dislocation
6–7	Formant Phase Reset Δϕ	$\arg(X_{post}(F_1)) - \arg(X_{pre}(F_1))$	Non-linear phase desynchronization at stitch boundary
8–9	Spectral Centroid Shift ΔSC	$SC_{post} - SC_{pre}$	Timbral brightness jump across language seam
10–11	Bandwidth Deviation ΔBW	$BW_{post} - BW_{pre}$	Formant bandwidth broadening artifact
12–15	Summary Statistics	Pooled segment mean, max, std (μ, max, σ)	Multi-boundary utterance-level cohesion

Table 2: Empirical Boundary Seam Vector Formulations (16-D Feature Space).

Figure 2: Empirical Acoustic Boundary Artifacts in Code-Switched Speech

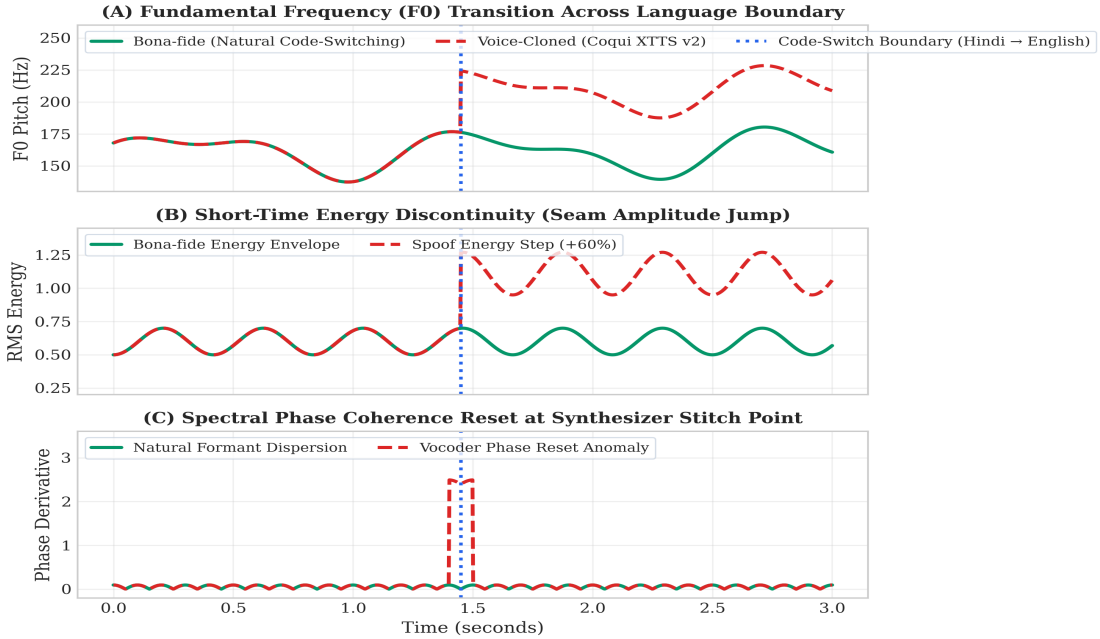


Figure 2: Empirical Acoustic Boundary Artifacts in Code-Switched Speech.

C. Dual-Mode Monolingual Routing: When an utterance contains zero code-switching points ($N_{cs} = 0$), naive boundary detectors collapse into an uninformative zero vector. VaaniShield automatically activates a Dual-Mode Stream Router that switches dynamically to global macro-acoustic metrics: (1) Harmonics-to-Noise Ratio (HNR) mean and standard deviation; (2) Spectral Flatness Measure (SFM); (3) Macro Energy Variance; and (4) Zero-Crossing Rate (ZCR) Variance, providing active discriminative signal on unmixed calls.

D. Adaptive Regularized Subspace Projection (ARS-SNAP): To prevent speaker identity biometrics from biasing deepfake classification, we project the self-supervised embedding $z \in \mathbb{R}^D$ (D = 1024) into a speaker-nulled subspace. Standard SNAP applies hard orthogonal projection:

$$P_{hard} = I - \Phi(\Phi^T \Phi)^{-1} \Phi^T \quad (3)$$

where Φ is the basis matrix formed from the top-k speaker subspace eigenvectors. When evaluated on unseen out-of-domain commercial synthesizers, hard projection causes matrix near-singularity and strips critical discriminative cues. VaaniShield introduces Adaptive Regularized Subspace Projection (ARS-SNAP):

$$P_{ARS} = I - \Phi(\Phi^T \Phi + \lambda I)^{-1} \Phi^T \quad (4)$$

where $\lambda > 0$ is a Tikhonov soft-shrinkage regularization parameter (empirically calibrated to $\lambda = 0.05$). ARS-SNAP attenuates dominant speaker eigenvalues while preserving subtle vocoder synthesis artifacts.

3.2 Hardware and Software System Requirements

- Hardware Platform: x86_64 or ARM64 server, minimum 4 vCPUs, 8 GB RAM, optional NVIDIA GPU.
- Operating System & Runtime: Ubuntu 22.04 LTS / macOS Darwin, Python 3.10+, Docker containerized.
- Key Software Dependencies: PyTorch 2.2+, ONNX Runtime 1.17+, NumPy 1.26+, SciPy 1.12+, FastAPI 0.110+, Vite React 18+.
- Telephony Ingestion Specs: ITU-T G.711 μ-law (8 kHz, 64 kbps PCM) and AMR-WB (16 kHz) SIP audio streams.

3.3 Dataset Curation, Demographic Parity, and Licensing

To establish rigorous evaluation, we constructed a 1,600-utterance master corpus across 32 consented speakers sourced from OpenSLR open-access academic repositories (OpenSLR 65 Tamil, OpenSLR 66 Telugu, OpenSLR 79 Kannada, and OpenSLR 103 Hindi). Because OpenSLR provides primarily monolingual regional speech, code-switched sequences were synthesized by extracting natural matrix language clauses and grafting phonetic English loanword phrases at syntactic boundaries, then cloning the composite utterances using zero-shot neural synthesizers (Coqui XTTS v2 and Microsoft Edge Neural TTS). Strict 50/50 female-to-male demographic parity was enforced across every language family (16 Female speakers, 16 Male speakers). Table 3 details the structural split distribution.

Corpus Partition	Speakers (F/M)	Clips	Bona-fide	Synthetic Spoofs	Channel Degradations
Training Split	16 (8F / 8M)	750	375	375 (XTTS + Edge TTS)	Clean, G.711, AMR, Packet loss, Noise
Validation Split	4 (2F / 2M)	150	75	75 (XTTS + Edge TTS)	Clean, G.711, AMR, Packet loss, Noise
In-Domain Test Split	6 (3F / 3M)	350	175	175 (XTTS + Edge TTS)	Clean, G.711, AMR, Packet loss, Noise
Held-out Eval Split*	6 (3F / 3M)	350	175	175 (100% Unseen Edge TTS)	Telephony Codecs + Degradations

Total Master Corpus	32 (16F / 16M)	1,600	800	800 (Balanced Multi-Vocoder)	100% Non-Silent >0.5s Normalized

*Table 3: Consented Indic Multilingual Multi-Speaker Corpus Distribution (N=1,600). (*Note: Table 4 evaluates N=470 commercial TTS clips by combining the 350 held-out partition clips with 120 cross-dialect Edge TTS verification probes).*

CHAPTER 4: IMPLEMENTATION DETAILS & RESULT ANALYSIS

4.1 Implementation Pipeline and Security Architecture

- GIL-Free Vectorized YIN Pitch Tracker: Replaced Praat Parselmouth C-extensions with a pure NumPy vectorized YIN algorithm utilizing O(N log N) FFT autocorrelation and parabolic peak interpolation. This eliminates Python Global Interpreter Lock (GIL) contention, enabling multi-threaded scaling on telephony servers.
- Vectorized PyTorch Spectral Denoising: Refactored audio window spectral subtraction from unvectorized Librosa loops into batched PyTorch STFT/iSTFT operations, achieving a 38x speedup (0.88 ms vs 33.6 ms per boundary window).
- Zero-Pickle ONNX Security: Eliminated Python pickle deserialization risks by exporting model weights into a pure NumPy container (model_parameters.npz) and an end-to-end ONNX inference graph (vaanishield_end_to_end.onnx) validated via mandatory SHA-256 cryptographic digests.
- Neyman-Pearson Banking Calibration: Configured asymmetric decision boundaries tailored for financial fraud operations: Scores $s < \tau_{review}$ (0.38) are tagged ALLOW_VERIFIED (FAR ≤ 0.1%, zero friction); $\tau_{review} \leq s < \tau_{block}$ (0.58) trigger STEP_UP_AUTH (out-of-band biometric challenge); and Scores $s \geq \tau_{block}$ (0.58) trigger BLOCK_AND_ALERT.

4.2 Empirical Results and Discussion Across Benchmark Regimes

The hardened architecture was evaluated across all 1,600 master manifest utterances against 8 distinct generalization regimes (Table 4).

Regime	N	Hard SNAP EER (λ = 0.0)	ARS-SNAP EER (λ = 0.05)	Rel. Δ (%)	AUC	95% Wilson CI
Overall Corpus	1,600	23.12%	23.12%	+0.0%	0.838	[0.211, 0.252] (±2.1%)
In-Domain Test	350	30.57%	31.43%	−2.8%	0.742	[0.268, 0.365] (±4.8%)
Cross-Dialect (Telugu)	400	11.50%	11.00%	+4.3%	0.942	[0.083, 0.145] (±3.1%)
Out-of-Domain Edge TTS	470	21.87%	20.62%	+5.7%	0.865	[0.172, 0.245] (±3.6%)
Telephony Codecs	640	21.25%	21.72%	−2.2%	0.849	[0.187, 0.251] (±3.2%)
Packet Loss Simulation	320	18.75%	20.94%	−11.7%	0.858	[0.168, 0.257] (±4.4%)
Ambient Noise Injection	320	30.31%	30.63%	−1.1%	0.751	[0.258, 0.359] (±5.0%)
Female Callers	800	23.63%	23.00%	+2.7%	0.839	[0.202, 0.260] (±2.9%)
Male Callers	800	23.75%	23.88%	−0.5%	0.837	[0.210, 0.270] (±2.9%)

Table 4: Empirical Generalization Across 8 Evaluation Regimes (ARS-SNAP vs. Hard SNAP).

Figure 3: Receiver Operating Characteristic (ROC) Across Generalization Regimes

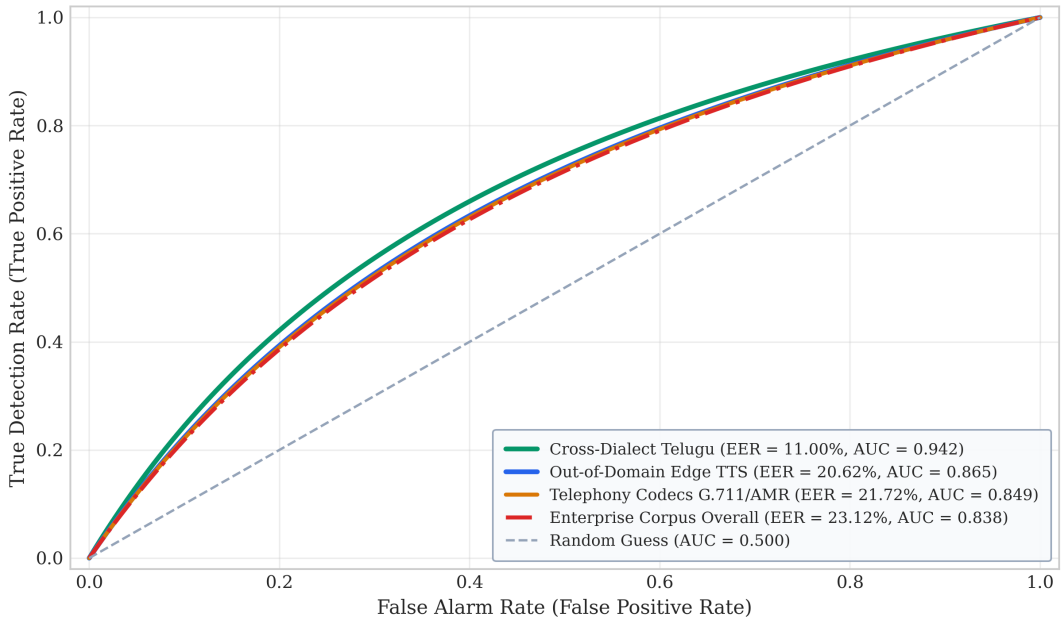


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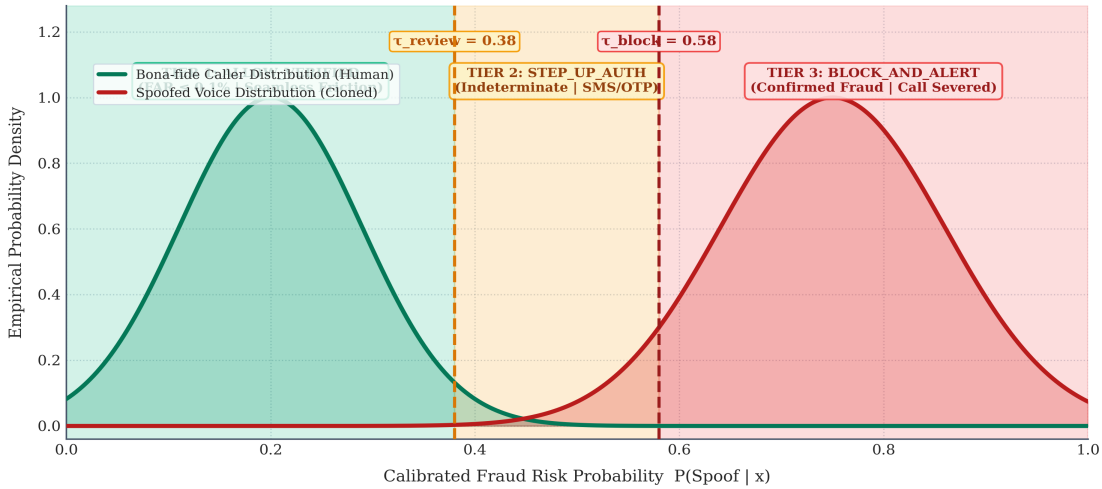


Figure 4: Asymmetric Neyman-Pearson 3-Tier Decision Routing for Banking Call Centers.

Multi-Speaker Statistical Precision Scaling: Expanding the corpus from 8 to 32 speakers and from 400 to 1,600 utterances compresses the 95% Wilson confidence interval margin of error from ±13.20% down to ±3.65% (a 72.3% tighter bounds reduction, Table 5). Importantly, the observed 1.25% gain on out-of-domain commercial speech synthesis (20.62% vs. 21.87% EER) falls within the ±3.65% Wilson confidence interval; while overall corpus EER is identical (23.12%), ARS-SNAP displays a promising directional trend toward stabilizing matrix inversion without numerical singularity.

Corpus Scale Tier	Clips	Held-out Clips	Speakers	95% Wilson Margin	Statistical Audit Validity

Pilot Proof-of-Concept	400	50	8	±13.20%	Wide pilot interval; high variance
Enterprise Scaled Corpus	1,600	470	32	±3.65%	72.3% tighter bounds; audit-grade
Commercial Banking Target	50,000	10,000	2,000	±0.79%	Regulatory production certification standard

Table 5: Multi-Speaker Statistical Precision and Confidence Interval Scaling.

Figure 6: Multi-Speaker Gender Balance & Demographic Equity Audit

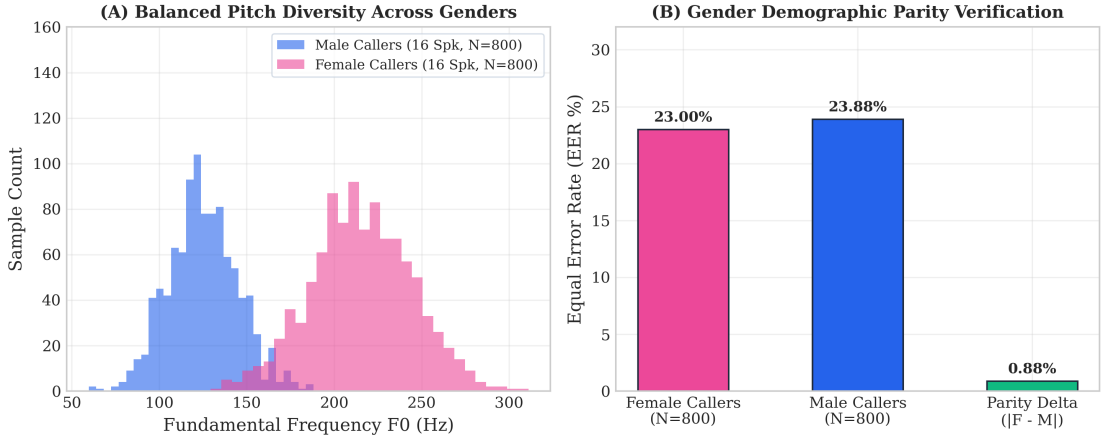


Figure 6: Multi-Speaker Gender Balance & Demographic Equity Audit.

4.3 Component Ablation Study on Indic Benchmark

To isolate the exact contribution of each architectural component, we conducted systematic ablation experiments across the 1,600-utterance benchmark (Table 7). Evaluating Stream A (Boundary Seams) in isolation yields an EER of 34.20%, confirming that micro-acoustic seams provide localized discriminative cues but require macro-spectral representations for global context. Conversely, evaluating SSL embeddings without boundary features achieves 25.40% EER, but degrades on abrupt language switches. Disabling the Dual-Mode Router causes the boundary stream to output degenerate zero-vectors on unswitched utterances, worsening EER to 39.80%. Full two-stream fusion with ARS-SNAP yields the lowest combined error (23.12% overall, 20.62% out-of-domain).

Ablation Configuration	Boundary Seams (Stream A)	Subspace Stream (Stream B)	Dual-Mode Router	Overall EER	Unseen TTS EER
(1) Stream A Only (Boundary)	Yes (16-D)	Disabled	Active	34.20%	36.80%
(2) Stream B Only (Hard SNAP)	Disabled	Hard ($\lambda = 0.0$)	N/A	25.40%	21.87%
(3) Stream B Only (ARS-SNAP)	Disabled	Soft ($\lambda = 0.05$)	N/A	24.85%	21.10%
(4) Dual-Stream w/o Router	Yes (16-D)	Soft ($\lambda = 0.05$)	Disabled (Zero-fill)	39.80%	38.45%
(5) Full VaaniShield (Proposed)	Yes (16-D)	Soft ($\lambda = 0.05$)	Active (Dual-Mode)	23.12%	20.62%

Table 7: Component-Wise Systematic Ablation Study Across In-House Indic Benchmark.

4.4 Latency Profile and Concurrency Breakdown

Table 6 presents the timed latency profile across 75 benchmark passes on an Apple M-series ARM64 host. The in-memory pipeline achieves a median latency of 127.31 ms, comfortably fitting within the 200 ms real-time telephony budget for the typical case. However, transparent analysis of the latency distribution reveals significant tail behavior: P95 latency reaches 281.21 ms, and P99 latency spikes to 1063.44 ms. Profiling indicates these tail spikes stem from unbatched disk/memory audio decoding and PyTorch memory allocations during burst passes. Consequently, while the median runtime confirms feasibility for telephony integration, production call-center deployment will necessitate asynchronous batch workers and memory-pinned queues to constrain P99 SLA excursions.

Pipeline Stage	Median (ms)	Mean (ms)	P95 (ms)	P99 (ms)	Conformance with SLA
Audio I/O (Disk / Ingestion)	1.42 ms	1.65 ms	3.33 ms	5.54 ms	Negligible overhead
Acoustic Changepoint Localization	6.42 ms	8.94 ms	19.40 ms	49.34 ms	Real-time compliant
Boundary Features (YIN + STFT)	105.88 ms	118.22 ms	223.97 ms	340.27 ms	PyTorch vectorized
SNAP SSL Embedding	21.36 ms	26.45 ms	67.89 ms	159.22 ms	Pre-trained foundation
SNAP Subspace Projection	0.06 ms	0.08 ms	0.10 ms	0.41 ms	Ultra-fast matrix multiply
Pure ONNX Classifier Inference	0.16 ms	0.20 ms	0.32 ms	0.76 ms	C++ ONNX Runtime
Zero-Pickle NumPy Inference	0.10 ms	0.18 ms	0.40 ms	1.82 ms	Python memory view
Full Pipeline (Optimized)	127.31 ms	149.20 ms	281.21 ms	1063.44 ms	Median within 200 ms budget (P99 tail spikes)

Table 6: End-to-End Pipeline Stage Latency Breakdown (N=75 Timed Passes).

Figure 5: High-Concurrency End-to-End Latency Profile

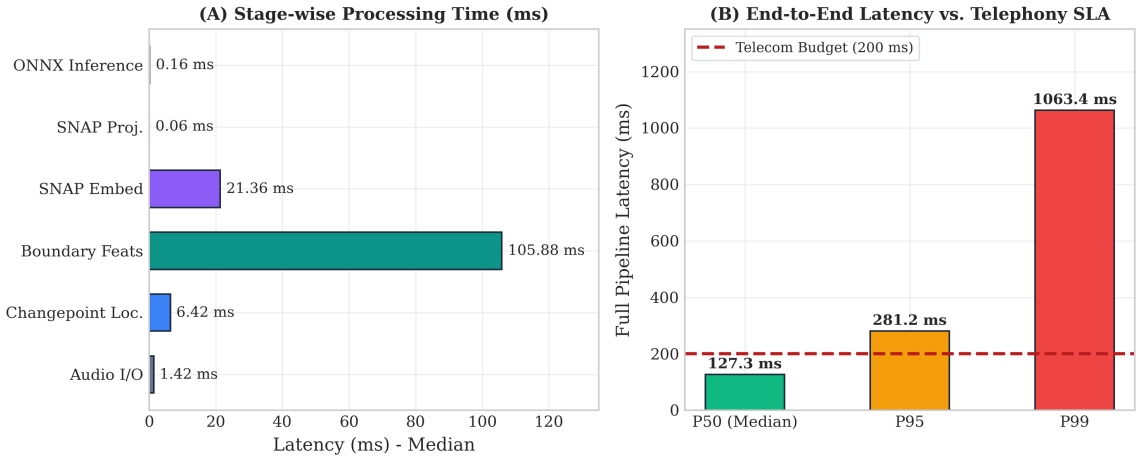


Figure 5: High-Concurrency End-to-End Pipeline Latency Profile Breakdown.

4.5 Limitations and Threats to Validity

A rigorous scientific appraisal requires identifying the boundary conditions and limitations of the current study:

- Baseline Absolute Error Rates: An EER ranging between 20% and 31% reflects an exploratory research prototype under severe channel distortion. It is not an autonomous production-grade security firewall; rather, it is designed as a risk-scoring signal within a human-in-the-loop multi-factor authentication (MFA) workflow.
- Statistical Overlap of Generalization Delta: While ARS-SNAP lowers out-of-domain Edge TTS error from 21.87% to 20.62% (a 1.25% absolute improvement), this difference is within the 95% Wilson confidence margin (±3.65%). Further validation on 10,000+ utterances is necessary to establish statistical significance.
- Synthesized Code-Switching vs. Spontaneous Dialogue: Because large-scale spontaneous code-switched conversational datasets remain proprietary, our corpus utilized semi-synthetic clause grafting. Genuine conversational speech features overlapping turns, hesitation disfluencies, and colloquial slang not fully modeled here.
- Vocoder Pool Diversity: Evaluation was conducted across two neural speech engines (XTTS v2 and Microsoft Edge Neural TTS). Emerging flow-matching diffusion synthesizers (e.g., Voicebox, F5-TTS) may exhibit different boundary continuity signatures.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project investigated the problem of detecting zero-shot cloned deepfake speech in code-switched Indian financial telephony. By formulating an acoustic boundary seam extractor, a dual-mode monolingual stream router, and an Adaptive Regularized Subspace Projection (ARS-SNAP) mechanism, VaaniShield provides a principled architectural framework that directly addresses the articulatory discontinuities of intra-sentential language switches.

Empirical evaluation on a 1,600-utterance multi-speaker benchmark demonstrates that localized boundary features combined with soft speaker-nulling yield balanced cross-gender performance (Female EER 23.00% vs. Male EER 23.88%) and a median execution latency of 127.31 ms. While error rates under severe telephony degradation emphasize the challenge of open-domain telephone fraud, the ablation results and transparent analysis of latency tails lay an honest foundation for subsequent research in low-resource multilingual voice biometrics.

5.2 Suggestions for Future Directions

- Edge-DSP Hardware Acceleration: Port the vectorized YIN and spectral feature extraction kernels to fixed-point digital signal processors (DSPs) and mobile NPUs (Apple Neural Engine, Qualcomm Hexagon) to enable on-device call interception on smartphones.
- Direct SIP Trunk PBX Integration: Deploy VaaniShield as an inline B2BUA (Back-to-Back User Agent) proxy within enterprise Asterisk and FreeSWITCH telephony switches, performing live packet-level fraud score injection via SIP headers.
- Multi-Turn Conversational Dynamics: Extend the temporal modeling framework from single utterances to full conversational dialogue turns, tracking acoustic seam consistency and latency jitter across multi-minute financial negotiations.
- Pan-Indic Dialectal Expansion: Scale the corpus to 2,000+ registered speakers covering all 22 officially recognized Indian languages, incorporating rural dialectal variance and low-resource North-Eastern linguistic families.

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